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1 MAPRO Full Analysis Test Run

Command executed:

```
bash ../scripts/driver.sh \  
-n ../data/neidb \  
< ../data/test_list.txt
```

The driver script processes each organism listed in `test_list.txt` independently.

1.1 1. Aquifex aeolicus VF5 (aae)

Output:

```
Analyzing aae Aquifex aeolicus VF5  
Collecting 1 genome record  
Downloading: tdata.zip  
...  
Error: There are no genome assemblies that match your query.
```

The type strain was found in the Neighbors database. Only one target genome accession was available. NCBI Datasets attempted to download the genome. No complete/chromosome-level assembly was available. As a result, no genome FASTA file was downloaded.

```
ls: cannot access 'aae/targets/'  
ls: cannot access 'aae/neighbors'
```

Since no genomes were available, MAPRO could not build a phylogeny and therefore skipped marker discovery for this species.

1.2 2. *Agrobacterium fabrum* C58 (afa)

Output:

```
Collecting 20 genome records
Collecting 68 genome records
```

20 genomes were identified as target genomes. 68 genomes were identified as neighboring genomes.

The genomes were downloaded and rehydrated successfully.

A phylogeny was constructed and a representative target genome was selected:

```
using tGCF_003667945.1_ASM366794v1_genomic.fna as target representative
```

Marker discovery (Fur):

Step	Sequences	Length	Ns
Subtraction_1	5305	1487141	0
Intersection	2188	469635	18938
Subtraction_2	55	39423	26398

Interpretation:

Subtraction₁: sequences present in target genomes but absent from neighbors. Intersection: sequences shared among all target genomes. Subtraction₂: final candidate marker regions.

1.3 General Workflow

For each organism, MAPRO performs:

Identify target and neighboring taxa using the Neighbors database. Download genomes from NCBI. Build a phylogeny. Select a representative target genome. Create a Fur database. Run Fur marker discovery. Produce marker sequences. Design primers from the discovered markers.